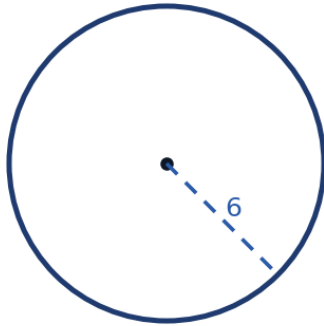


Circles & Coordinate Geometry



The equation of a circle

- 1 Write the equation of a circle with center $(-4, 7)$ and radius 9.
- 2 The circle below has the radius shown. Find its area and circumference, in terms of π .



- 3 A circle has diameter 26 and is centered at the origin. Write its equation, and state its radius.

Completing the square

- 4 Find the center and radius of $x^2 + y^2 - 6x + 8y - 11 = 0$.
- 5 Find the center and radius of $x^2 + y^2 + 10x - 4y + 13 = 0$.

Distance, midpoint, and slope

- 6 Find the distance between $(-3, 5)$ and $(4, -7)$, to the nearest hundredth.
- 7 Find the midpoint and slope of the segment joining $(2, -6)$ and $(-8, 10)$.
- 8 Find the slope of the line through $(-2, 7)$ and $(6, -5)$, and state whether the line rises or falls from left to right.

Coordinate proofs

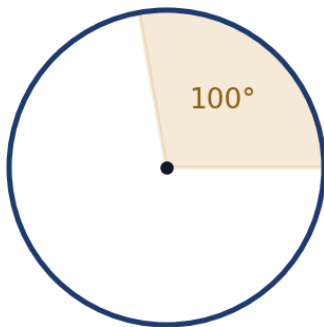
- 9 Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle, using slopes.
- 10 Show that $A(1, 2)$, $B(4, 3)$, $C(3, 6)$, $D(0, 5)$ form a parallelogram, by showing both pairs of opposite sides are parallel.
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Tangent lines to circles

- 11 A circle has center $O(2, 3)$ and radius 5. Point $P(2, 12)$ lies outside the circle. Find the length of the tangent from P to the circle, to the nearest hundredth.
- 12 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ on the circle.
- 13 A circle has center $(0, 0)$ and radius 10. Determine whether the point $(6, 9)$ lies inside, on, or outside the circle.
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Circle theorems

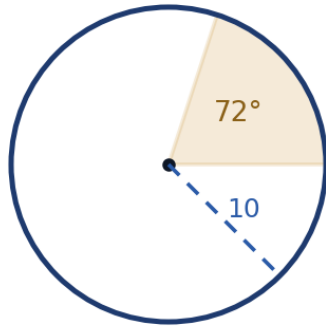
- 14 A central angle of 84° subtends an arc. Find the measure of the inscribed angle subtending the same arc.
- 15 A chord subtends the central angle shown below. Find the measure of the inscribed angle subtending the same (minor) arc.



Arc length and sector area

- 16 A circle has radius 12 cm. Find the arc length subtended by a central angle of $\frac{5\pi}{6}$ radians.

- 17 The circle below has the radius and shaded sector shown. Find the area of the sector, in terms of π .



- 18 A sector of a circle with radius 9 has area 27π . Find the central angle, in radians.

Conics: ellipses and hyperbolas

- 19 Identify the type of conic and state its center for $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$.
- 20 Identify the type of conic, its center, and its asymptote slopes for $\frac{(x+3)^2}{4} - \frac{y^2}{9} = 1$.

A well-designed word problem: a Ferris wheel

- 21 This question has three parts. A circular Ferris wheel has a diameter of 40 m, with its center 22 m above the ground. Set up a coordinate system with the origin at ground level, directly below the center. (a) Write the equation of the circle traced by a passenger cabin. (b) A support cable runs from a ground anchor at (26, 0) to a point of tangency on the wheel. Find the length of this cable, to the nearest hundredth of a meter. (c) Find the height above the ground of a cabin when it is 12 m horizontally from the vertical line through the center.